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↳ Happens Why?

↳ The generator discovers that a particular fake sample fools discr. successfully,

↳ therefore it keeps producing same successful sample repeatedly.

" FEDERATED LEARNING "

→ A distributed Machine Learning approach where multiple devices or organizations collaboratively train a model without sharing their raw data

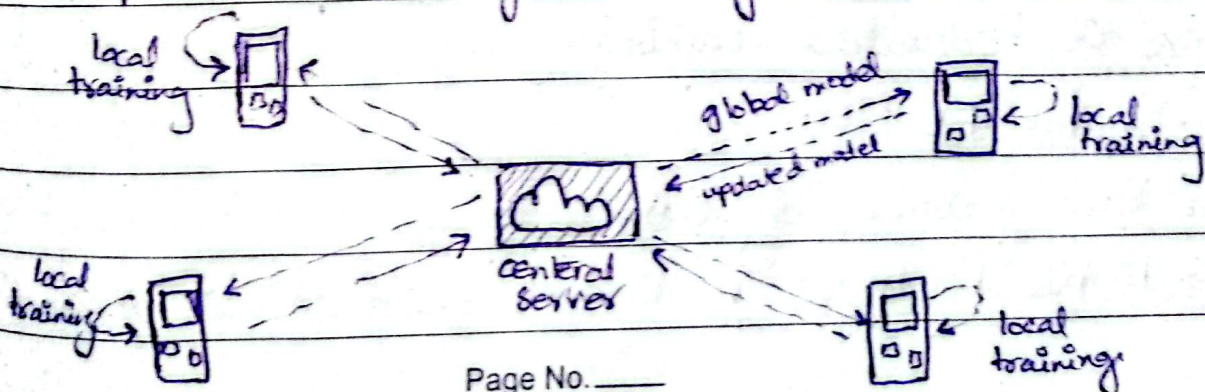
→ Instead of sharing data to a central server:

↳ devices keeps data locally

↳ Only model updates / params are shared

↳ Updated model sent back from central server to local device

→ Improves → privacy, security and data ownership



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Centralized Learning

- All data sent to one server
- Simple Optimization — full data visible
- Privacy Risk : data breach exposes everything
- Regulatory issues

Federated Learning

- Data stays on device — only gradients sent
- Complex Optimization — heterogeneous data
- Reduces Privacy risk — no raw data transfer
- Compliant with data locality laws

• Advantages

- ↳ Privacy Reservation
- ↳ Reduce Data Transfer
- ↳ Compliance Friendly
- ↳ Decentralized learning

• Types of Federated Learning

① Horizontal FL :

- ↳ Same features, diff data instances

- ↳ Mobile keyboard predictions, smart device personalization

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② Vertical FL

- ↳ Same data instances, diff features
- ↳ Joint fraud detection, credit scoring

③ Federated Transfer Learning

- ↳ Diff features and diff samples
- ↳ Uses knowledge from a source domain to improve performance on a related but diff target domain
- ↳ Adapting a general medical model to a specific hospital's patient data

• Process Of FL

- ① Server creates a model, initializes global model param θ_0 and broadcast to all clients
- ② At each round t , server chooses C ^{out} of K total clients
- ③ Each of selected client runs E epochs on its local data D_k , starting from θ_t params, producing θ_t^k
- ④ Server aggregates client updated weights
- ⑤ Updated global model θ_{t+1} is broadcast back, repeated for T rounds.

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- Disadvantages

- ↳ Communication overhead
- ↳ Slow training
- ↳ High internet usage
- ↳ Network congestion
- ↳ Slow devices delay training rounds
- ↳ Some clients cannot train large models

- Federated Averaging

↳ The avg of the model (weight) updates sent by client to central server is done using a weighted average.

↳ Ex: Three clients involved in a FL

- $w_1 = [1.1, 0.9] \rightarrow 100$ samples
- $w_2 = [1.3, 1.0] \rightarrow 300$ "
- $w_3 = [1.2, 0.95] \rightarrow 100$ "

$$\begin{aligned}w_{\text{new}} &= \frac{100 \cdot [1.1, 0.9] + 300 \cdot [1.3, 1.0] + 100 \cdot [1.2, 0.95]}{100 + 300 + 100} \\&= \frac{[110, 90] + [390, 300] + [120, 95]}{500}\end{aligned}$$

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$$W_{\text{new}} = [1.24, 0.97]$$

- Reduces Privacy Concerns

- ① Adds carefully calibrated random noise that hides exact info about individual samples
- ② Encrypt updates before sending them.